OBSTACLE AVOIDING ROBOT

A PROJECT REPORT

Submitted by

ASHIRBAD BISWAL (240301120168)

SUMIT KUMAR PRUSTY (240301120140)

SHUBHANKAR TRIPATHY (240301120165)

ISWAR BEHERA (240301120142)

CHINMAYA MOHAPATRA (240301120151)

in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Centurion
UNIVERSITY
Shaping Lives...
Empowering Communities...

SCHOOL OF ENGINEERING AND TECHNOLOGY

BHUBANESWAR CAMPUS

CENTURION UNIVERSITY OF TECHNOLOGY AND MANAGEMENT

ODISHA

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CERTIFICATE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF ENGINEERING AND TECHNOLOGY

BHUBANESWAR CAMPUS

BONAFIDE CERTIFICATE

Certified that this project report **"OBSTACLE AVOIDING ROBOT"** is the Bonafide work of the team consisting of **SHUBHANKAR TRIPATHY** carried out the project work under my supervision. This is to further certify to the best of my knowledge, that this project has not been carried out earlier in this institute and the university.

SIGNATURE

(Prof. Chinmayee Dora)

Head of the Department of ECE

Certified that the above-mentioned project has been duly carried out as per the norms of the college and statutes of the university.

SIGNATURE

(Prof. Rajkumar Mohanto)

HEAD OF THE DEPARTMENT / DEAN OF THE SCHOOL

Professor of Computer Science and Engineering

DEPARTMENT SEAL

DECLARATION

I hereby declare that the project entitled **"OBSTACLE AVOIDING ROBOT"** submitted for the "Minor Project" of 4th semester B. Tech in Computer Science and Engineering is my original work and the project has not formed the basis for the award of any Degree / Diploma or any other similar titles in any other University / Institute.

Name of the Students:	**Registration No:**
ASHIRBAD BISWAL	240301120168
SUMIT KUMAR PRUSTY	240301120140
SHUBHANKAR TRIPATHY	240301120165
ISWAR BEHERA	240301120142
CHINMAYA MOHAPATRA	240301120151

Place: BHUBANESWAR

Date:

NAME:

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CHINMAYA MOHAPATRA	240301120151

Place: BHUBANESWAR

Date:

ABSTRACT

The project titled **"OBSTACLE AVOIDING ROBOT"** focuses on designing and developing an autonomous robotic system capable of detecting and avoiding obstacles using sensor-based technology. The main objective of the project is to create an intelligent robot that can navigate its path without human intervention by using an ESP32 microcontroller and ultrasonic sensors.

The robot is built using hardware components such as ESP32, ultrasonic sensors, motor drivers, and DC motors. The ultrasonic sensor detects obstacles by measuring the distance between the robot and surrounding objects. The ESP32 processes the sensor data and controls the movement of the robot accordingly. The programming is carried out using the Arduino IDE, which provides an easy and efficient platform for coding and uploading instructions to the microcontroller.

The system works by continuously monitoring the environment. When an obstacle is detected within a specific range, the robot automatically stops and changes its direction to avoid collision. This ensures smooth and safe navigation in different environments. The use of ESP32 enhances the performance of the system by providing faster processing, low power consumption, and wireless communication capabilities.

This project demonstrates the practical implementation of robotics, embedded systems, and automation technologies. It provides hands-on experience in hardware integration, programming, and real-time decision-making. The obstacle avoiding robot can be applied in various fields such as autonomous vehicles, industrial automation, surveillance systems, and smart robotics applications. The system successfully showcases efficiency, reliability, and the importance of intelligent robotic systems in modern technology.

This streamlined setup enhances efficiency for developers and small businesses seeking quick web presence.

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INTRODUCTION

In today's modern era, ROBOTICS has become an essential technology in automation and intelligent systems. It enables machines to perform tasks efficiently with minimal human intervention. One of the key features of autonomous systems is the ability to detect and avoid obstacles, ensuring safe and reliable operation in real-world environments.

The project titled "OBSTACLE AVOIDING ROBOT" aims to design and develop an autonomous robotic system that can navigate its surroundings without human control. The robot is built using an ESP32 MICROCONTROLLER, ULTRASONIC SENSOR, motor driver, and DC motors. The programming is done using the ARDUINO IDE, which provides an easy platform for coding and controlling the robot's operations.

The system works by continuously monitoring the environment using the ultrasonic sensor, which measures the distance between the robot and nearby objects. Based on this data, the ESP32 processes the input and controls the movement of the robot. When an obstacle is detected, the robot automatically STOPS AND CHANGES DIRECTION, ensuring safe navigation.

This project helps in understanding the core concepts of EMBEDDED SYSTEMS, AUTOMATION, and real-time decision-making. It also provides practical knowledge of hardware integration and programming. The obstacle avoiding robot can be applied in areas such as AUTONOMOUS VEHICLES, industrial automation, and smart surveillance systems.

Thus, this project demonstrates how intelligent robotic systems can improve efficiency, safety, and performance in modern technological applications.

PROBLEM STATEMENT

In today's rapidly advancing technological world, **ROBOTICS** and **AUTOMATION** are becoming essential in various fields such as industry, healthcare, transportation, and surveillance. The need for **AUTONOMOUS SYSTEMS** that can operate without human intervention has increased significantly. One of the most important requirements of such systems is the ability to navigate safely in an environment filled with obstacles.

Traditional robotic systems often rely on **MANUAL CONTROL**, which limits their efficiency and increases the chances of human error. These systems require continuous monitoring and are not suitable for applications where real-time response is critical. In dynamic environments, where obstacles may appear unexpectedly, manual control becomes ineffective and may lead to **COLLISIONS** and system damage.

The existing robotic systems without proper sensing mechanisms suffer from several limitations such as **LACK OF INTELLIGENCE**, poor adaptability, and inability to respond to changing surroundings. These systems cannot make decisions independently, which restricts their use in advanced applications like **AUTONOMOUS VEHICLES**, robotic navigation, and industrial automation. Therefore, there is a need to develop intelligent robots that can sense their environment and take decisions accordingly.

The core problem addressed in this project is the development of a robot capable of **OBSTACLE DETECTION AND AVOIDANCE** in real-time. For a robot to move safely, it must continuously monitor its surroundings, calculate distances, and respond immediately when an obstacle is detected. This requires the integration of **SENSORS**, **MICROCONTROLLERS**, and efficient programming techniques. To overcome these challenges, the proposed system uses an **ESP32 MICROCONTROLLER** along with an **ULTRASONIC**

SENSOR to detect obstacles. The ultrasonic sensor works on the principle of sound wave reflection, measuring the distance between the robot and nearby objects. The ESP32 processes this data and controls the robot's movement through motor drivers. The programming is implemented using the **ARDUINO IDE**, which allows easy development and real-time control.

The project focuses on solving the following key challenges:

1. **LACK OF AUTOMATION**
   Most traditional robots require human control, which reduces efficiency and limits real-world applications.
2. **COLLISION AND SAFETY ISSUES**
   Without proper sensing, robots may collide with obstacles, leading to hardware damage and system failure.
3. **REAL-TIME PROCESSING**
   The robot must process sensor data instantly and take quick decisions to avoid obstacles.
4. **ADAPTABILITY TO ENVIRONMENT**
   The system should work effectively in different environments with varying obstacle positions.
5. **COST-EFFECTIVE IMPLEMENTATION**
   The solution should be affordable and easy to build using commonly available components like ESP32 and sensors.
6. **RELIABILITY AND PERFORMANCE**
   The robot should provide consistent and accurate performance during continuous operation.

Thus, the problem statement of this project is to design and develop an **INTELLIGENT OBSTACLE AVOIDING ROBOT** that can perform **REAL-TIME OBSTACLE DETECTION**, make decisions autonomously, and ensure **SAFE AND EFFICIENT NAVIGATION** using embedded system technology.

This project aims to bridge the gap between basic robotic systems and advanced intelligent automation by creating a system that is reliable, efficient, and capable of operating independently in real-world environments.

OBJECTIVES

The main objective of this project is to design and develop an **OBSTACLE AVOIDING ROBOT** using an **ESP32 MICROCONTROLLER** and **ULTRASONIC SENSOR**, capable of detecting and avoiding obstacles automatically. The project aims to demonstrate how **ROBOTICS**, **EMBEDDED SYSTEMS**, and **AUTOMATION TECHNOLOGIES** can be integrated to create an intelligent and efficient robotic system.

This project not only focuses on hardware implementation but also emphasizes learning the core principles of **SENSOR INTEGRATION**, **REAL-TIME PROCESSING**, and **ROBOT CONTROL SYSTEMS**. The objectives are categorized into technical, functional, and educational goals, as outlined below:

A. Technical Objectives

1. **To design and develop an obstacle avoiding robotic system:**
   Create a fully functional robot that can detect obstacles and navigate safely without human intervention.

2. **To implement ESP32-based control system:**
   Utilize the **ESP32 MICROCONTROLLER** for processing sensor data and controlling motor operations efficiently.

3. **To integrate ultrasonic sensor for distance measurement:**
   Use the **ULTRASONIC SENSOR** to measure distance and detect obstacles based on sound wave reflection.

4. **To develop motor control mechanism:**
   Interface motor drivers and DC motors to control the movement of the robot in different directions.

5. **To program the system using Arduino IDE:**
   Write and upload code using the **ARDUINO IDE** for real-time control and decision-making.

B. Functional Objectives

1. **To enable automatic obstacle detection and avoidance:**
   Ensure the robot can **DETECT OBSTACLES** and **CHANGE DIRECTION AUTOMATICALLY** without manual control.
2. **To ensure smooth and efficient navigation:**
   Design the system to move continuously while avoiding collisions in different environments.
3. **To achieve real-time response:**
   Process sensor data instantly to provide quick and accurate movement decisions.
4. **To ensure reliability and performance:**
   Develop a system that performs consistently under different working conditions.
5. **To implement a cost-effective robotic solution:**
   Use easily available and affordable components for building the system.

C. Educational and Learning Objectives

1. **To gain practical knowledge in robotics:**
   Understand the working principles of **ROBOTICS SYSTEMS** and autonomous navigation.

2. **To apply embedded system concepts:**
   Implement concepts like **SENSOR INTERFACING**, **MICROCONTROLLER PROGRAMMING**, and control logic.
3. **To develop hardware and software integration skills:**
   Learn how to connect and synchronize hardware components with software instructions.
4. **To enhance problem-solving abilities:**
   Identify challenges in robot movement and develop efficient solutions for obstacle avoidance.

The successful completion of this project helps in understanding the complete process of designing and developing an **AUTONOMOUS ROBOTIC SYSTEM**. It bridges the gap between theoretical knowledge and practical implementation by applying concepts of **EMBEDDED SYSTEMS**, **AUTOMATION**, and **REAL-TIME CONTROL**.The project's objectives not only enhance technical skills but also provide a strong foundation for advanced applications in **SMART ROBOTICS**, **AUTONOMOUS VEHICLES**, and **INDUSTRIAL AUTOMATION**.

PROPOSED SYSTEM

The proposed system aims to design and implement an **AUTONOMOUS OBSTACLE AVOIDING ROBOT** that can navigate its environment safely without human intervention. The robot is developed using an **ESP32 MICROCONTROLLER** for processing, an **ULTRASONIC SENSOR** for obstacle detection, and motor drivers with DC motors for movement control. The entire system is programmed using the **ARDUINO IDE**, ensuring efficient and real-time operation.

The system overcomes the limitations of traditional manually controlled robots by providing **AUTOMATIC NAVIGATION**, **REAL-TIME DECISION MAKING**, and improved safety. The integration of sensors and microcontroller enables the robot to detect obstacles and respond instantly, making it suitable for dynamic environments.

The obstacle avoiding robot focuses on three key modules — **SENSING MODULE**, **CONTROL MODULE**, and **ACTUATION MODULE**. These modules work together to achieve smooth and intelligent movement.

1. Sensing Module

The sensing module is responsible for detecting obstacles in the robot's path. It uses an **ULTRASONIC SENSOR** which works on the principle of sound wave reflection.

- The sensor sends ultrasonic waves and receives the reflected signals.
- It calculates the **DISTANCE** between the robot and nearby objects.
- Provides real-time data to the microcontroller for processing.

This module ensures accurate and continuous monitoring of the environment.

2. Control Module

The control module is the brain of the system, implemented using the **ESP32 MICROCONTROLLER**.

• It processes the data received from the ultrasonic sensor.
• Executes programmed instructions written in the **ARDUINO IDE**.
• Makes decisions based on distance values (e.g., stop, turn, or move forward).
• Ensures **REAL-TIME RESPONSE** and efficient system control.

The ESP32 enhances system performance with faster processing and reliability.

3. Actuation Module

The actuation module is responsible for the movement of the robot.

• Uses **MOTOR DRIVER (L298N)** to control DC motors.
• Converts control signals into physical movement.
• Enables the robot to move forward, backward, left, and right.

This module ensures smooth and controlled navigation of the robot.

4. Working Principle

The robot continuously monitors its surroundings using the ultrasonic sensor. When an obstacle is detected within a predefined distance:

1. The sensor sends distance data to the ESP32.
2. The ESP32 processes the data in real-time.
3. The robot automatically **STOPS AND CHANGES DIRECTION**.
4. It selects an alternate path and continues movement. This ensures **SAFE AND COLLISION-FREE NAVIGATION**.

5. Implementation and Programming

The system is programmed using the **ARDUINO IDE**, where the logic for obstacle detection and movement control is written.

• Sensor interfacing and data processing are handled in code.

• Motor control instructions are executed based on conditions.

• The system runs continuously in a loop for real-time operation.

6. Advantages of the Proposed System

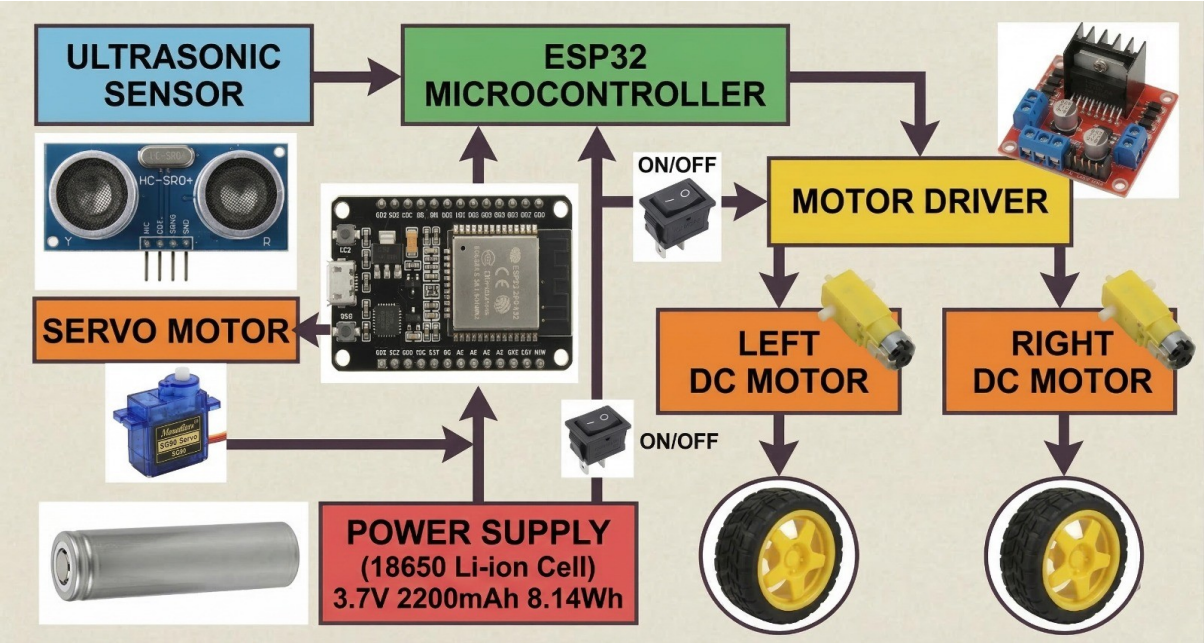
• **AUTOMATIC OPERATION:** No need for manual control

• **REAL-TIME RESPONSE:** Immediate action on obstacle detection

• **HIGH EFFICIENCY:** Smooth and continuous movement

• **COST-EFFECTIVE:** Uses affordable and easily available components

• **RELIABILITY:** Performs consistently in different environments



FEASIBILITY

Operational Feasibility

Operational feasibility focuses on how well the proposed system will function in a real-world environment and whether it will effectively meet the required needs. The proposed system — **OBSTACLE AVOIDING ROBOT** — is designed to operate autonomously and navigate safely by detecting and avoiding obstacles in its path.

The system provides a simple and efficient robotic solution that can function without continuous human intervention. It uses an **ULTRASONIC SENSOR** to detect obstacles and an **ESP32 MICROCONTROLLER** to process the data and control movement. The robot responds instantly to environmental changes, ensuring **SAFE AND SMOOTH NAVIGATION**.

This system replaces traditional manually controlled robots with an **AUTOMATED SOLUTION** that improves efficiency and reduces human effort. It can be used in various real-world applications such as **AUTONOMOUS VEHICLES**, industrial automation, and surveillance systems.

Moreover, the robot is designed using simple and easily available components, making it easy to operate and maintain. The system performs continuously once powered, requiring minimal supervision.

Thus, the system is operationally feasible because it:

• Meets the functional requirement of **OBSTACLE DETECTION AND AVOIDANCE.**

• Provides **AUTOMATIC NAVIGATION** without human control
• Operates efficiently in real-time environments
• Reduces human effort and improves safety
• Ensures reliable and consistent performance

Technical Feasibility:

Technical feasibility evaluates whether the required technology for developing and implementing the system is available and suitable.

The system is built using an **ESP32 MICROCONTROLLER**, which is powerful, cost-effective, and suitable for real-time applications. The **ULTRASONIC SENSOR** is used for accurate distance measurement based on sound wave reflection. The robot also uses a **MOTOR DRIVER (L298N)** and **DC motors** to control movement. The **SG90 Servo Motor** in an obstacle-avoiding robot rotates the ultrasonic sensor to scan surroundings for obstacles.

The programming is done using the **ARDUINO IDE**, which is widely used for embedded system development due to its simplicity and flexibility. It allows easy coding, testing, and uploading of programs to the microcontroller.

All the hardware components used in this project are easily available, affordable, and compatible with each other. The system does not require complex infrastructure or advanced tools, making it suitable for student-level implementation.

Thus, the project is technically feasible because:

• Uses **EASILY AVAILABLE AND LOW-COST COMPONENTS**
• Supports **REAL-TIME PROCESSING** using ESP32
• Simple programming using **ARDUINO IDE**

- Easy to design, implement, and maintain
- Provides reliable performance with minimal complexity

METHODOLOGY

In this project, we have adopted a systematic and iterative approach for the development of the **OBSTACLE AVOIDING ROBOT**. The methodology focuses on step-by-step design, testing, and improvement of the system to ensure efficient performance. The development process includes **DESIGN, IMPLEMENTATION, TESTING, AND OPTIMIZATION**, allowing continuous verification at each stage.

The reason behind selecting this approach is its flexibility and simplicity, which allows modifications and improvements during development. The system is built, tested, corrected, and re-tested multiple times to ensure accuracy and reliability. This approach helps in identifying errors early and improving the overall performance of the robot.

The methodology involves both **HARDWARE IMPLEMENTATION** and **SOFTWARE DEVELOPMENT**, which are integrated to achieve the desired output.

1. Hardware Design Module

This module focuses on assembling all the physical components required for the robot.

• Selection of components such as **ESP32 MICROCONTROLLER**, **ULTRASONIC SENSOR**, **MOTOR DRIVER (L298N)**, **SG90 Servo Motor**, **BO Motor** & **DC MOTORS**
• Designing the robot chassis and mounting all components properly
• Establishing connections between sensor, microcontroller, and motors
• Ensuring proper **POWER SUPPLY** using batteries

This module forms the physical structure of the robot.

2. Sensor and Detection Module

This module is responsible for **OBSTACLE DETECTION**.

• The **ULTRASONIC SENSOR** emits sound waves and receives reflected signals
• Calculates the **DISTANCE** between the robot and obstacles
• Sends real-time data to the ESP32 for processing

This ensures continuous monitoring of the environment.

3. Control and Processing Module

This module acts as the brain of the system.

• The **ESP32 MICROCONTROLLER** processes the sensor data
• Executes decision-making logic based on programmed conditions
• Determines actions such as **MOVE FORWARD**, **STOP**, or **TURN**

This module ensures **REAL-TIME DECISION MAKING** and control.

4. Programming Module

The system is programmed using the **ARDUINO IDE**.

• Writing code for sensor interfacing and motor control
• Implementing conditions for obstacle detection
• Uploading and testing the program on ESP32
• Debugging and optimizing the code for better performance

5. Testing and Validation Module

This module ensures the proper functioning of the system.

• Testing the robot in different environments
• Checking accuracy of **DISTANCE MEASUREMENT**

• Verifying obstacle avoidance behaviour.
• Making necessary corrections and improvements

6. Working Process

The working of the system follows a continuous cycle:

1. The ultrasonic sensor detects obstacles
2. Distance data is sent to the ESP32
3. The microcontroller processes the data
4. The robot automatically **STOPS AND CHANGES DIRECTION**
5. The robot continues moving in a safe path

This methodology ensures that the **OBSTACLE AVOIDING ROBOT** is developed in a structured and efficient manner. It combines **HARDWARE INTEGRATION**, **SOFTWARE PROGRAMMING**, and **REAL-TIME TESTING** to achieve reliable and accurate performance.

IMPLEMENTATION

The implementation phase is one of the most crucial stages in the development life cycle of a project. It is the phase where the system design is transformed into a working model that can be executed, tested, and used in real-world applications. The implementation of the **OBSTACLE AVOIDING ROBOT** involves both **HARDWARE ASSEMBLY** and **SOFTWARE PROGRAMMING**, ensuring proper integration of all components.

This project integrates multiple technologies such as the **ESP32 MICROCONTROLLER**, **ULTRASONIC SENSOR**, **MOTOR DRIVER (L298N)**, and **DC MOTORS**. The system is programmed using the **ARDUINO IDE**, which provides a simple and efficient platform for writing and uploading code to the microcontroller.

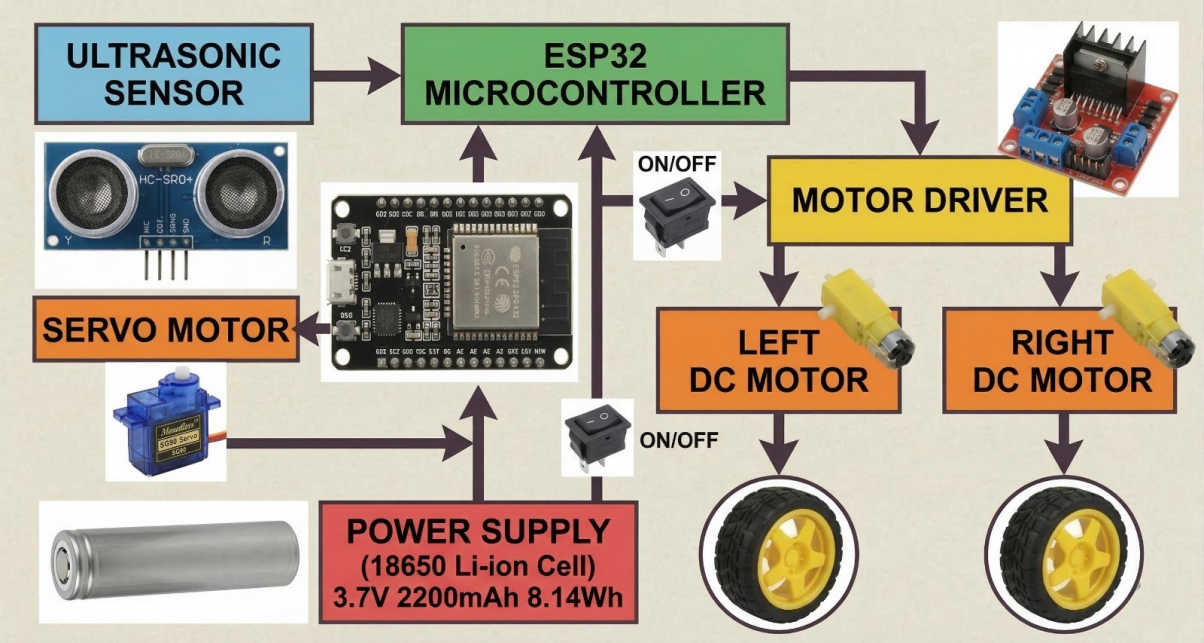
The implementation process includes assembling the hardware components, establishing proper connections, programming the control logic, and testing the robot in different environments to ensure smooth and efficient operation. The robot is designed to detect obstacles and automatically change its direction, ensuring **SAFE AND COLLISION-FREE MOVEMENT**.

1. Components Used

The following components are used in the development of the project:

- ESP32 Microcontroller
- Ultrasonic Sensor (HC-SR04)
- Motor Driver (L298N)
- DC Motors
- Robot Chassis

- Battery
- Jumper Wires
- SG90 Servo Motor



Components Used in Obstacle Avoiding Robot

2. Working Principle

The working of the robot is based on **ULTRASONIC SENSING** and **REAL-TIME PROCESSING**. The ultrasonic sensor continuously emits sound waves and receives the reflected signals from nearby objects. Based on the time taken for the signal to return, the **DISTANCE** between the robot and the obstacle is calculated.

The ESP32 processes this data and decides the movement of the robot. When an obstacle is detected within a predefined range, the robot automatically **STOPS AND CHANGES DIRECTION** to avoid collision. Otherwise, it continues to move forward.

3. Code Implementation

The system is programmed using the **ARDUINO IDE**, where the logic for obstacle detection and motor control is implemented.

- Sensor data is read and processed continuously
- Conditional statements are used for decision-making
- Motor control signals are generated based on distance

Project_Obstacle.ino

```
#include <ESP32Servo.h>
#define trigPin 5
#define echoPin 18
#define IN1 26
#define IN2 27
#define IN3 25
#define IN4 33
#define servoPin 13
Servo myServo;
long duration;
int distance;
int rightDistance, leftDistance;
int getDistance() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH, 30000);
  if (duration == 0) {
    return 999;
  }
  distance = (duration * 0.0343) / 2;
  return distance;
}
```

```
void moveForward() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void moveBackward() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

void stopCar() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
}
```

```
void turnLeft() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void turnRight() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  myServo.attach(servoPin);
  myServo.write(90);
  delay(500);
```

```
  Serial.begin(9600);
}
void loop() {
  int frontDistance = getDistance();
  if (frontDistance < 20) {
    stopCar();
    delay(300);
    // Go backward
    moveBackward();
    delay(500);
    stopCar();
    delay(300);
    // Look left
    myServo.write(0);
    delay(500);
    leftDistance = getDistance();
    // Look right
    myServo.write(180);
    delay(500);
```

RESULT ANALYSIS

The Result Analysis section evaluates the overall performance, functionality, and efficiency of the developed system after successful implementation. It focuses on how well the **OBSTACLE AVOIDING ROBOT** meets its objectives and performs under real-world conditions.

The robot, developed using an **ESP32 MICROCONTROLLER** and **ULTRASONIC SENSOR**, was tested under different conditions to analyze its performance. The evaluation was based on parameters such as **OBSTACLE DETECTION ACCURACY**, **RESPONSE TIME**, **MOVEMENT EFFICIENCY**, and **RELIABILITY**. The results show that the system operates effectively as an autonomous robotic system capable of safe navigation.

During testing, the robot was able to detect obstacles within a specific range and respond accordingly. When an obstacle was detected, the robot successfully **STOPPED AND CHANGED DIRECTION**, avoiding collisions in most cases. The **ULTRASONIC SENSOR** provided accurate distance measurements, and the ESP32 processed the data efficiently to control the robot's movement.

The system demonstrated good **REAL-TIME RESPONSE**, as the robot reacted quickly to obstacles without noticeable delay. The movement of the robot was smooth, and it was able to navigate through different paths without manual intervention. The overall **PERFORMANCE AND STABILITY** of the system were satisfactory.

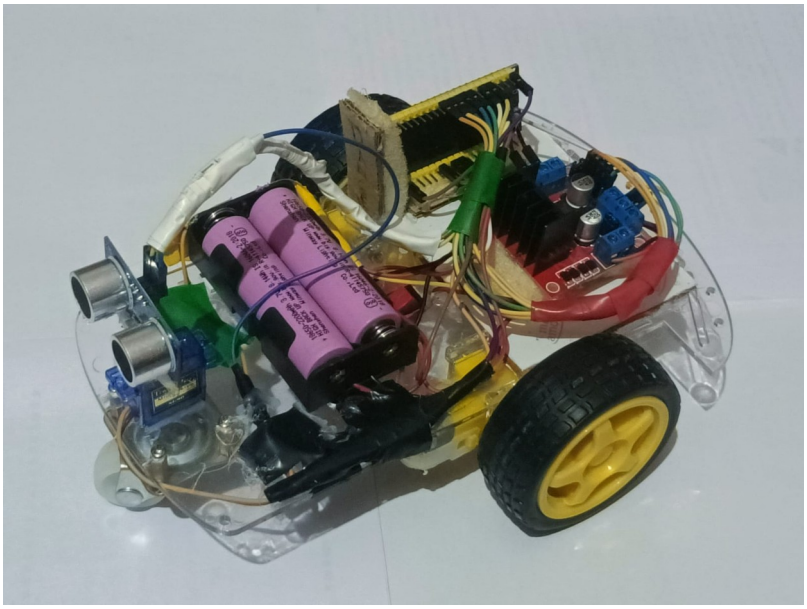
Various test conditions were applied, such as placing obstacles at different distances and angles. The robot performed consistently in most scenarios, although minor variations in detection accuracy were observed due to environmental factors like surface type and object shape.

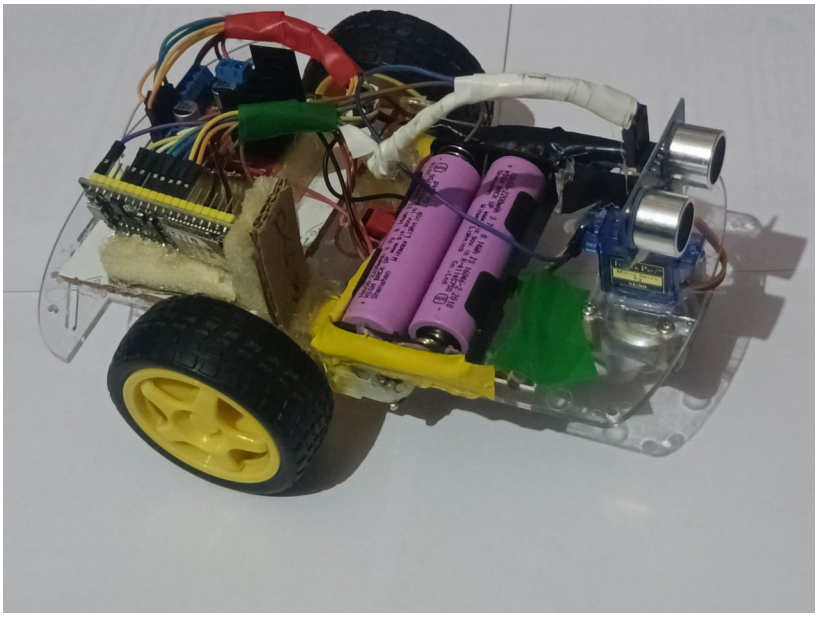
The analysis also highlights that the system is **COST-EFFECTIVE**, **ENERGY-EFFICIENT**, and easy to implement using basic components. The integration of

hardware and software was successful, and the robot operated continuously without major issues.

Thus, the result analysis confirms that the **OBSTACLE AVOIDING ROBOT** meets the desired objectives and provides an efficient solution for **AUTONOMOUS NAVIGATION**. The system can be further enhanced by adding advanced sensors or improving the control algorithm for better accuracy and performance.

PROJECT IMAGES / SAMPLE OUTPUTS





CONCLUSION

The project titled **"OBSTACLE AVOIDING ROBOT"** was developed with the primary objective of designing and implementing an autonomous robotic system capable of detecting and avoiding obstacles using sensor-based technology. The system successfully demonstrates the integration of **ROBOTICS**, **EMBEDDED SYSTEMS**, and **AUTOMATION TECHNOLOGIES** to create an efficient and intelligent robot.

The project shows how components like the **ESP32 MICROCONTROLLER**, **ULTRASONIC SENSOR**, motor driver, and DC motors can work together to build a reliable and cost-effective robotic system. The use of the **ARDUINO IDE** made it easier to program and control the robot, ensuring smooth operation and real-time responsiveness.

During the implementation and testing phase, the robot achieved its key objectives — including **AUTOMATIC OBSTACLE DETECTION**, **REAL-TIME DECISION MAKING**, and **SAFE NAVIGATION**. The system was able to detect obstacles accurately and respond by stopping and changing direction, thereby avoiding collisions effectively. The overall performance of the robot was stable and reliable under different test conditions.

Through this project, practical knowledge was gained in **HARDWARE INTEGRATION**, **MICROCONTROLLER PROGRAMMING**, and **SENSOR INTERFACING**. It also provided hands-on experience in designing and developing a complete robotic system from scratch. This enhances understanding of real-world applications of robotics and automation.

Additionally, this project highlights how simple and affordable components can be used to develop intelligent systems capable of performing complex tasks. The **OBSTACLE AVOIDING ROBOT** can be further improved by incorporating

advanced technologies such as **ARTIFICIAL INTELLIGENCE**, **VISION SYSTEMS**, or additional sensors for enhanced performance and functionality.

Overall, the project successfully achieves its objectives and demonstrates the potential of **AUTONOMOUS ROBOTIC SYSTEMS** in improving efficiency, safety, and performance in modern technological applications.

REFERENCES

- **Arduino Official Documentation — Arduino IDE & Programming Guide**
  https://www.arduino.cc/en/Guide
- **ESP32 Documentation — ESP32 Microcontroller Reference**
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- **Ultrasonic Sensor (HC-SR04) Datasheet — Working and Specifications**
  https://components101.com/sensors/ultrasonic-sensor-working-pinout-datasheet
- **L298N Motor Driver Module — Technical Guide**
  https://components101.com/modules/l298n-motor-driver-module
- **GeeksforGeeks — Arduino and Sensor Interfacing Tutorials**
  https://www.geeksforgeeks.org/
- **Tutorials Point — Robotics and Embedded Systems Basics**
  https://www.tutorialspoint.com/robotics/index.htm
- **Electronics Hub — Obstacle Avoiding Robot using Arduino/ESP32**
  https://www.electronicshub.org/
- **Circuit Digest — Obstacle Avoiding Robot Projects and Guides**
  https://circuitdigest.com/
- **All About Circuits — Electronics and Embedded System Learning**
  https://www.allaboutcircuits.com/
- **Stack Overflow — Arduino and ESP32 Programming Discussions**
  https://stackoverflow.com/
- **YouTube — Practical Tutorials on Obstacle Avoiding Robot**
  https://www.youtube.com/
- **Research Papers on Robotics and Automation — IEEE Xplore**
  https://ieeexplore.ieee.org/

ASSESSMENT:

Internal:

SL NO	RUBRICS	FULL MARK	MARKS OBTAINED	REMARKS
1	Understanding the relevance, scope and dimension of the project	10		
2	Methodology	10		
3	Quality of Analysis and Results	10		
4	Interpretations and Conclusions	10		
5	Report	10		
	Total	**50**		

Date: **Signature of the Faculty**

COURSE OUTCOME (COs) ATTAINMENT

➢ **Expected Course Outcomes (COs):**

(Refer to COs Statement in the Syllabus)

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➢ **Course Outcome Attained:**

How would you rate your learning of the subject based on the specified COs?

☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

LOW *HIGH*

➢ **Learning Gap (if any):**

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➢ **Books / Manuals Referred:**

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Date: **Signature of the Student**

➢ **Suggestions / Recommendations:**

(By the Course Faculty)

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Date: **Signature of the Faculty**